

Answers learning task 13

2026 – 2027

Exercise 8.4: A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 25$.

- (a) Find a 95 % CI for μ when $n = 10$ and $\bar{x} = 1000$.
- (b) Find a 95 % CI for μ when $n = 25$ and $\bar{x} = 1000$.
- (c) Find a 99 % CI for μ when $n = 10$ and $\bar{x} = 1000$.
- (d) Find a 99 % CI for μ when $n = 25$ and $\bar{x} = 1000$.
- (e) How does the length of the CIs computed above change with the changes in sample size and confidence level?

Answer

- (a) [984.5051, 1015.4949]
- (b) [990.2002, 1009.7998]
- (c) [979.6363, 1020.3637]
- (d) [987.1209, 1012.8791]
- (e) Increasing the sample size n decreases the length of the CI, whereas increasing the confidence level increases the length of the CI.

Exercise 8.7: Consider the gain estimation problem in Exercise 8.4.

- (a) How large must n be if the length of the 95 % CI is to be 30?
- (b) How large must n be if the length of the 99 % CI is to be 30?

Answer

- (a) $n \geq 11$
- (b) $n \geq 19$

Exercise 8.9: Suppose that $n = 100$ random samples of water from a freshwater lake were taken and the calcium concentration (mg/L) measured. A 95 % CI on the mean calcium concentration is $0.52 \leq \mu \leq 0.74$.

- (a) Would a 99 % CI calculated from the same sample data be longer or shorter?
- (b) Consider the following statement: “There is a 95 % chance that μ is between 0.52 and 0.74.” Is this statement correct? Explain your answer.
- (c) Consider the following statement: “If $n = 100$ random samples of water from the lake

were taken and the 95 % CI on μ computed, and this process were repeated 1000 times, 950 of the CIs would contain the true value of μ .” Is this statement correct? Explain your answer.

Answer

- (a) Longer
- (b) No, when the construction of a 95 %-confidence interval for μ is repeated a large number of times, the true mean will belong to the calculated interval in 95 % of the cases.
- (c) Yes, see explanation part (b).

Exercise 8.11: The yield of a chemical process is being studied. From previous experience, yield is known to be normally distributed and $\sigma = 3$. The past five days of plant operation have resulted in the following percent yields: 91.6, 87.34, 90.8, 89.65, and 91.3. Find a 95 % two-sided confidence interval on the true mean yield.

Answer

[87.5084, 92.7676]

Exercise 8.17: The life in hours of a 75-watt light bulb is known to be normally distributed with $\sigma = 20$ hours. Suppose you wanted the total width of the two-sided confidence interval on mean life to be six hours at 95 % confidence. What sample size should be used?

Answer

$n \geq 171$

Exercise 8.29: A postmix beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{x} = 30$ mL. Suppose the standard deviation is known and equal to $\sigma = 0.5$ mL. Find a 95 % CI on the mean volume of syrup dispensed.

Answer

[29.8040, 30.1960]

Exercise 7.63: Let $f(x) = (1/\theta)x^{(1-\theta)/\theta}$, $0 < x < 1$, and $0 < \theta < \infty$. Show that $\hat{\theta} = -(1/n) \sum_{i=1}^n \ln(X_i)$ is the maximum likelihood estimator for θ and that $\hat{\theta}$ is an unbiased estimator for θ .

Answer

The likelihood function is $L(\theta) = \theta^{-n} \prod_{i=1}^n x_i^{(1-\theta)/\theta}$. Taking the natural log gives:

$$\ln L(\theta) = -n \ln(\theta) + \frac{1-\theta}{\theta} \sum_{i=1}^n \ln(x_i)$$

Setting the derivative with respect to θ equal to zero:

$$\frac{d \ln L(\theta)}{d\theta} = -\frac{n}{\theta} - \frac{1}{\theta^2} \sum_{i=1}^n \ln(x_i) = 0 \implies \hat{\Theta} = -\frac{1}{n} \sum_{i=1}^n \ln(X_i)$$

To show it is unbiased, let $Y = -\ln(X)$. The CDF of Y for $y > 0$ is $P(-\ln(X) \leq y) = P(X \geq e^{-y}) = 1 - e^{-y/\theta}$, so $Y \sim \text{Exponential}(\theta)$ with $E(Y) = \theta$. Thus:

$$E(\hat{\Theta}) = E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = \frac{1}{n} \cdot n\theta = \theta$$

Hence, $\hat{\Theta}$ is an unbiased estimator for θ .